01. Software architecture Intro

Software Architectures
Master in Informatics Engineering

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Agenda

- Meaning of software architecture
- Software architecture in organizations
- Organization & IT - Meet TOGAF & COBIT

1.1 - What is software architecture?

- there are multiple definitions to it.
- Let’s agree on some views of it

Foundations

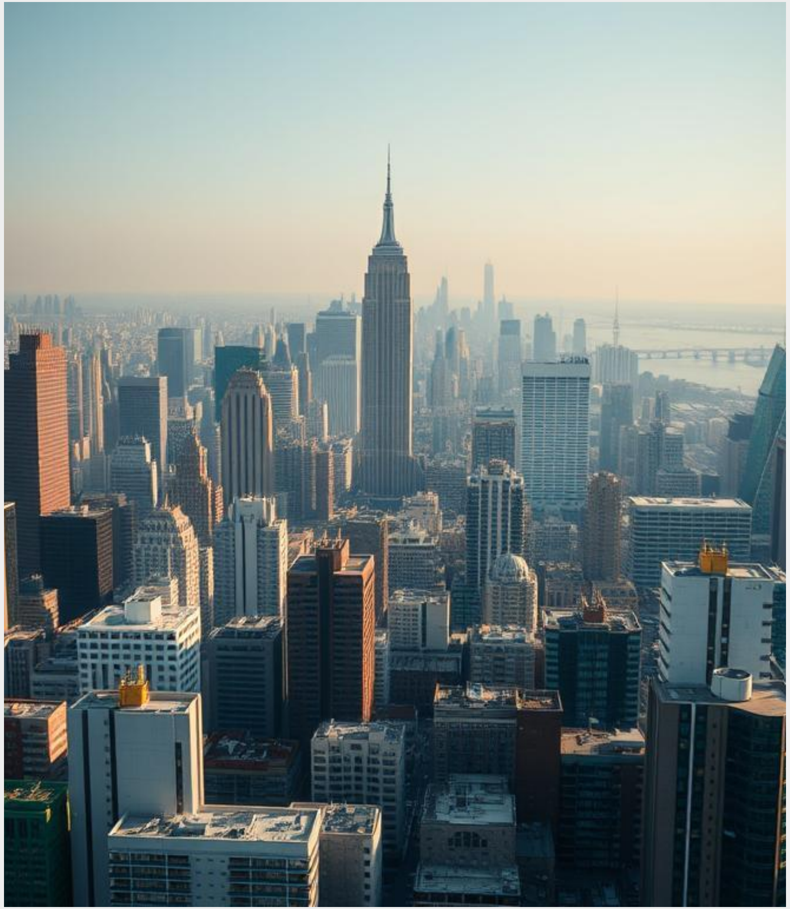
Understanding Software Architecture

Key Definitions and Perspectives

- Software architecture addresses the design beyond algorithms and data structures, focusing on the overall system structure.
- Garlan (1992) describes it as a new problem of conceiving, designing, and specifying system structure.
- Bass et al. (2003) define it as the structure of software components, their externally visible properties, and relationships.

As the size and complexity of software systems increases, the design problem goes beyond computing algorithms and data structures: ***conceiving, designing, and specifying the overall structure of the system emerges as a new kind of problem****… This is design at the level of software architecture. (Garlan, 1992)*

The software architecture of a program or computing system is the structure or structures of the system, encompassing ***software components****, the* ***externally visible properties*** *of these components, and* ***the relationships between them****. (Bass et all, 2003)*

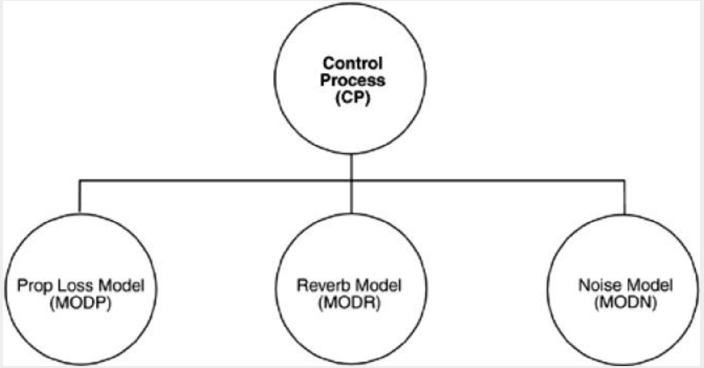


Architecture

Understanding the Diagram

Key Insights from the Diagram

- The diagram identifies four elements: CP, MODP, MODR, and MODN.
- Three elements (MODP, MODR, MODN) should exhibit greater similarity to each other than to CP.
- Each element maintains a type of relationship or link with the others, indicating interconnectedness.
- Since an architecture is a set of components and connections, this is the blueprint of an architecture.
- Is it?



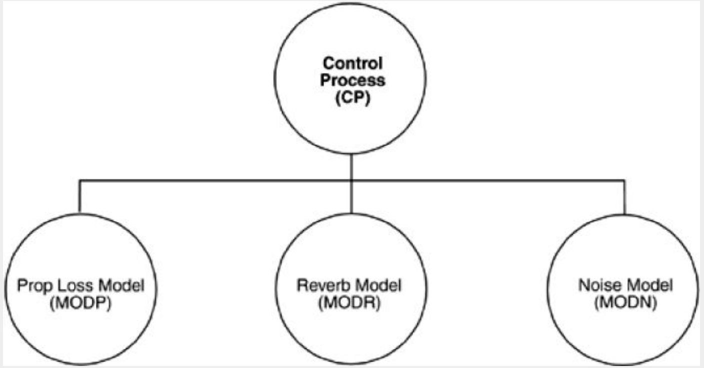
Top view of the architecture of a control system for an underwater acoustic simulation

Architecture

Understanding the Diagram

What does the diagram tells us?

- Are they distributed functions, processes, programs, or something else?
- What are their responsibilities?
- What do they do?
- What is their role in the system?
- What do the links mean?
- Do the links mean communication, control, sending information, invocation, synchronization, sharing of sensitive information, …?
- What are the communication mechanisms?
- What information flows through these mechanisms?

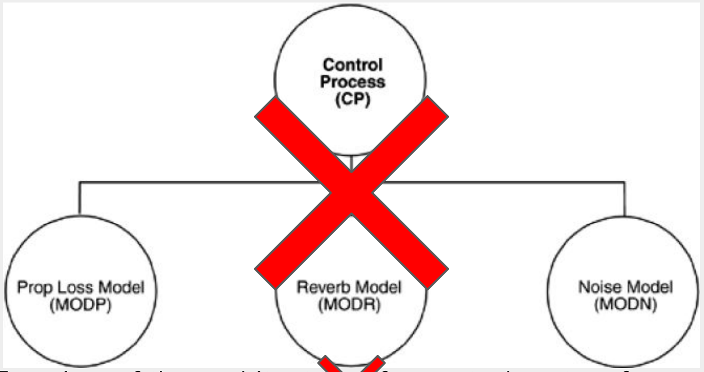


Top view of the architecture of a control system for an underwater acoustic simulation

Understanding the Diagram: Underwater Acoustic Simulation

Key Considerations in Diagram Interpretation

- The nature of the elements: Are they hardware components, software modules, or both?
- Significance of separation: Does the physical or logical separation impact system performance or function?
- Processor allocation: Do different elements operate on separate processors to enhance efficiency?
- Timing of operations: Are elements running synchronously or asynchronously, affecting task flow?
- Significance of CP's separate level: Does this imply a control precedence or is it an aesthetic choice due to space constraints?



Top view of the architecture of a control system for an underwater acoustic simulation

So … what is it then?

Architecture is concerned with the selection of architectural elements, their ***interaction****, the* ***constraints*** *and* ***rules*** *in these elements and their interactions... Design is related to the* ***modularization*** *and* ***detailed interfaces of architectural elements****, their algorithms and procedures, the types of data necessary to support the architecture and to satisfy its requirements. (Perry and Wolf, 1992)*

The architecture… ***<u>is specifically silent… on implementation details</u>*** *(algorithms and data structures). Architectural drawing uses a richer collection of abstractions than is achieved through Object Oriented Design (Monroe et al, 1997)*

Then let there be standards!
ISO/IEC/IEEE 42010:2022

- Software Architecture: *Fundamental concepts or properties of a system in its environment embodied in its elements, relationships, and in the principles of its design and evolution*
  - Emphasis is on capturing not only the static structure but also the **dynamic aspects** and **evolutionary considerations** of the system
- Interpretation:
  - Software architecture is a fundamental part of a software system
  - A software system is situated in an environment, and its software architectures takes into consideration the environment in which it must operate
  - An architecture description document the architecture and communicates to stakeholders how the architecture meets the system's needs
  - Architecture views are created from the architecture description, and each view covers one or more architecture concerns of the stakeholders

Why should we care about software architecture? Let's just code this!

- It is the software's blueprint, its foundation
- It stands as schematics based on decisions. That could probably not be possible without the schematics in the first place.
- It's iterative until the final drawings
- Every software has an architecture, so we might as well document it and produce it well!
- Architecture addresses and enables software to fulfill all requirements (functional, non-functional and operational)
    - No architecture, no requirements fulfillment (or a hard time on it, at least)!

Software is only as good as its architecture

- Software quality attributes are impacted by the architectural choices (deliberately or inadvertently)
  - such as… maintainability, interoperability, security and performance (we’ll get there during the semestre)
- Looking at the architecture, one can infer which qualities are being favoured and which are deemed as important
  - This saves time and money! If only by implementing it you can determine the qualities… then possible rework will be costly!

Direct effects of a good software architecture

- Ease the communication with stakeholders
  - can form the basis of discussions with the team
  - Its abstract nature means that it *should* be fairly understandable by everyone
- Helps on software changes
  - calculating its impact
  - reduces complexity of software, ergo changes will be simpler
- Provides patterns and reusable models
- Improves estimations for task completion
- Good intro material for newcomers

What about the software architect?

- Provides technical leadership into the project, management, customers and developers
- Responsible for the software architecture, its design and the architecture documentation
- Needs to understand the business domain
- Helps on analysing and gathering requirements
- Technical topics:
    - software architecture patterns (pros and cons)
    - how to handle cross cutting concerns
    - know how to integrate legacy applications
    - be able to design software that adapts to changes and evolves over time
- He is just a team player! can be influenced by the teams' performance and can impact the teams' performance.

Organisational goals significantly shape software architecture.

The organisation's goals, objectives, stakeholders, project management, and processes directly influence how software architecture is designed and implemented. This interplay ensures that software solutions are aligned with the organisational vision and objectives.

Architecture Roles

One Software Architect to Rule Them All?

Enterprise Architect

Focuses on the overall structure and strategy of an organisation's IT systems. Aligns IT solutions with business goals and ensures interoperability across departments.

Data Architect

Specialises in designing and managing an organisation's data architecture. Focuses on data models, databases, data flow, and data security.

Solution Architect

Works on the design and implementation of specific solutions or projects. Develops detailed specifications for how software components should be built and integrated.

Cloud Architect

Specialises in designing and implementing cloud-based solutions. Works with cloud technologies and services to optimise scalability, performance, and cost.

Application Architect

Concentrates on the architecture of a specific application or suite of applications. Defines how individual applications interact within the larger system.

Security Architect

Focuses on the security aspects of software and system architecture. Designs and implements security measures to protect against cyber threats.

Architecture

Role of Software Architects

Integration Architect

Ensures that different software systems can work seamlessly together. Designs solutions for integrating diverse applications and systems.

DevOps Architect

Integrates development and operations processes to streamline the software delivery pipeline. Focuses on automation, collaboration, and continuous improvement.

Mobile Architect

Specialises in designing software architectures for mobile applications. Considers constraints and opportunities specific to mobile platforms.

Big Data Architect

Specialises in designing architectures for handling and analysing large volumes of data. Works with technologies like Hadoop, Spark, and distributed databases.

UI/UX Architect

Focuses on the user interface (UI) and user experience (UX) aspects of software. Designs the overall look, feel, and usability of software applications.

IoT Architect

Designs architectures for systems involving interconnected devices. Addresses challenges related to data exchange, security, and scalability in IoT ecosystems.

Integration

Aligning software development with organisational goals is crucial.

Effective alignment is achieved through clear communication and understanding of both software architecture and the organisation's strategic objectives. Important architectural decisions and their implications must be explainable to the board to ensure alignment and support.

1.3 - Business alignment and the role of the software architect

Aligning IT strategies with business goals

IT Initiatives support and drive business objectives, leading to enhanced competitiveness, operational efficiency and innovation. Alignment enables...

01	**Improved Decision-Making**	Alignment ensures IT investments align with business priorities, enhancing decision-making.
02	**Increased Agility**	Organisations become more responsive to market changes by aligning IT with business strategies.
03	**Enhanced Innovation**	Partnership between IT and business fosters innovation through new technologies.
04	**Optimised Costs and Investments**	Alignment prioritises IT projects for efficient resource allocation based on business impact.
05	**Risk Management**	Aligning IT and business strategies aids in effective risk identification and mitigation.

Role of IT Governance and Architectural Frameworks

Architectural frameworks provide structure, processes and mechanisms to ensure success in multiple ways...

01	**Standardisation and Best Practices**	Frameworks like TOGAF and COBIT offer methodologies to reduce complexity and enhance efficiency.
02	**Strategic Alignment**	Aligns IT projects with business strategies, ensuring IT efforts deliver value.
03	**Risk Management and Compliance**	Ensures IT systems comply with regulations, managing information security risks.
04	**Performance Measurement**	Includes mechanisms for monitoring IT performance against business objectives.
05	**Resource Optimisation**	Helps optimise IT investments, allocating resources to high-value initiatives.

Understanding TOGAF & COBIT

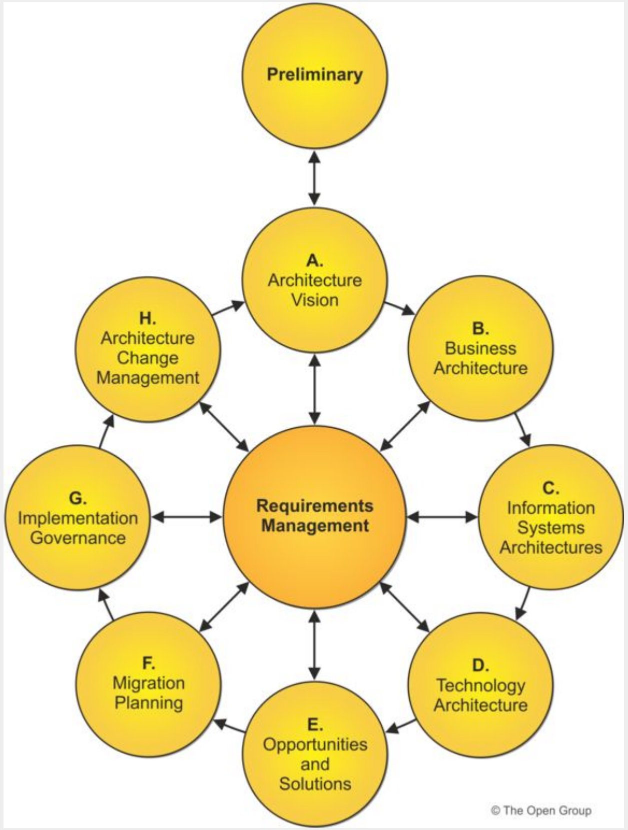
- TOGAF - The Open Group Architecture Framework focuses on providing a comprehensive approach to the design, planning, implementation, and governance of enterprise information architecture.
- COBIT - Control Objectives for Information and Related Technologies provides a framework for developing, implementing, monitoring, and improving IT governance and management practices.
- Both TOGAF and COBIT are high-level frameworks that help organisations align their IT strategies with business goals, ensuring effective governance and management of enterprise IT.
- TOGAF is often used to improve organisational efficiency by providing an architectural foundation, while COBIT ensures that IT systems are managed and controlled effectively.
- Together, these frameworks support enterprise architecture and IT governance, enabling organisations to achieve strategic objectives and improve performance.

Frameworks

Understanding TOGAF

Key Aspects of TOGAF

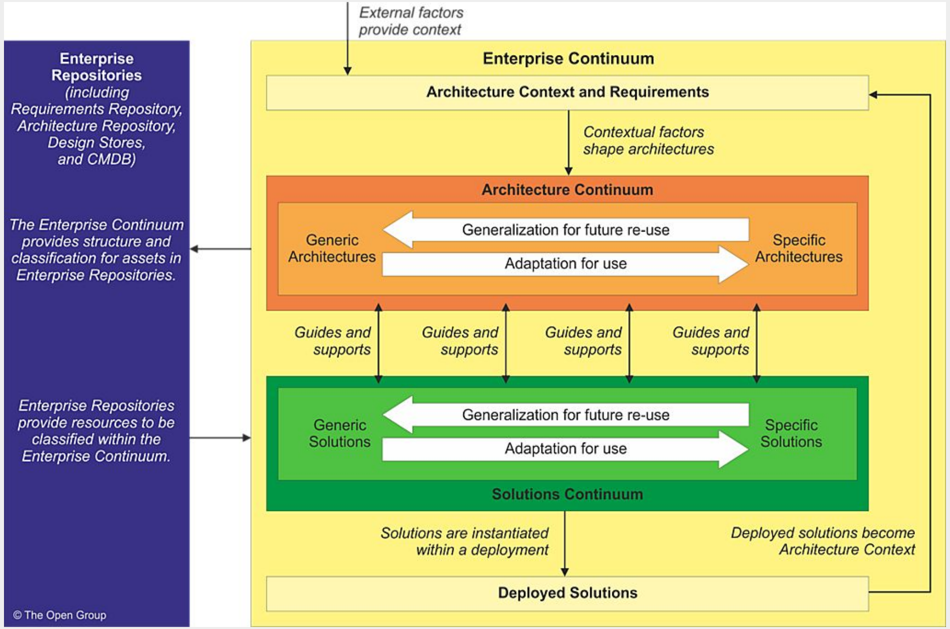
- It provides a comprehensive approach to designing, planning, implementing, and governing an enterprise IT architecture.
- TOGAF aims to align IT goals with overall business goals, ensuring that the architecture supports the organisation's objectives effectively.
- The framework includes the Architecture Development Method (ADM) for detailed architectural development processes.
- For more information, refer to the official TOGAF documentation at https://pubs.opengroup.org/architecture/togaf9-doc/arch/ .



TOGAF Architecture Development Method

Frameworks

Understanding the TOGAF Enterprise Continuum



- The Enterprise Continuum provides a framework and context to support relevant architecture assets in executing the Architecture Development Method (ADM).
- It serves as a comprehensive repository that categorises and manages architectural artefacts such as models, patterns, and other resources.
- The practical implementation of the Enterprise Continuum typically takes the form of an Architecture Repository, facilitating the organisation and retrieval of architectural information.
- The Continuum enables the integration and alignment of architecture with business strategy, ensuring consistent and strategic growth.
- It supports interoperability and standardisation across different architecture projects within an organisation.

Overview

More on Enterprise Continuum

Definition

The Enterprise Continuum is a framework in TOGAF for classifying enterprise architecture assets, aiding in structured management and reuse.

Levels of Abstraction

Includes Foundation, Common Systems, Industry, and Organisation-Specific levels, each building on the previous.

Supports Architecture Development

Aids in evolving architectures, encouraging reusability, and enhancing decision-making and interoperability.

Architecture Continuum

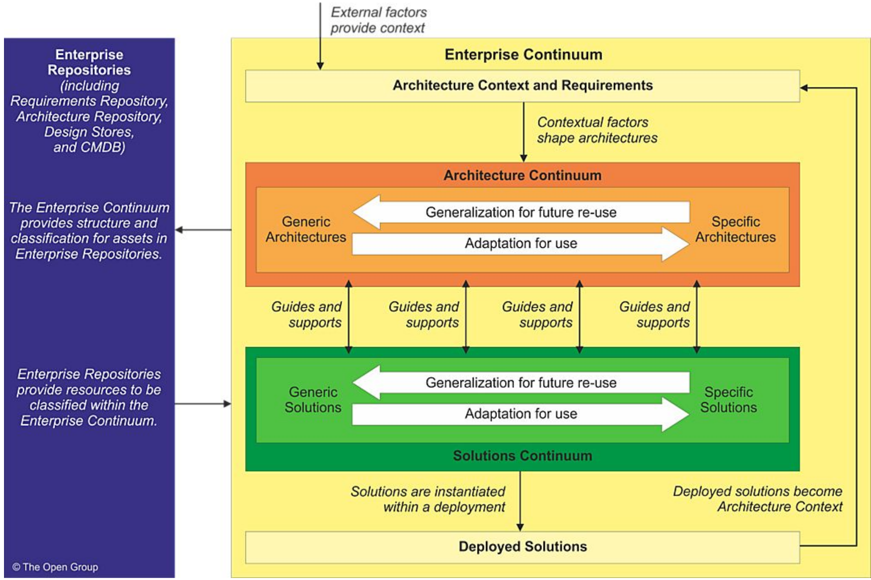
Organises architectures from abstract principles to specific solutions, refining them progressively for organisational needs.

Integration with TOGAF ADM

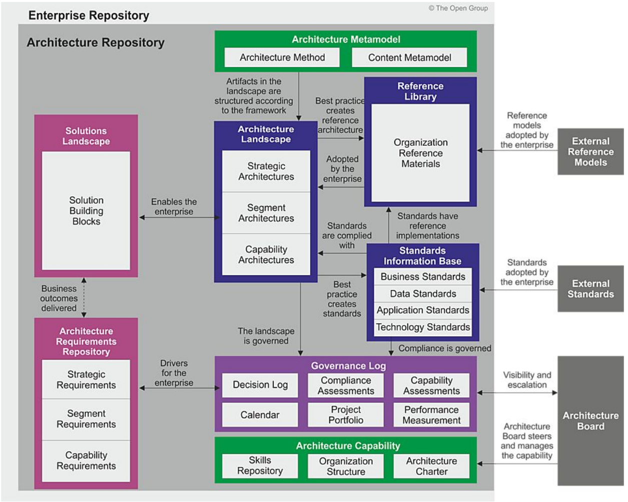
Provides a reference framework for architecture development, aligning with business objectives and IT strategy.

Solutions Continuum

Focuses on concrete implementations, guiding selection of technology components that align with architectural goals.



TOGAF Architecture Repository



Centralised Hub for Artefacts

The Architecture Repository serves as a centralised hub for all architectural artefacts within an enterprise, ensuring consistent access and management of architectural information.

Components of the Repository

This repository encompasses various components, including the Architecture Metamodel and a collection of architectural patterns and standards that guide development practices.

Organising Architectural Outputs

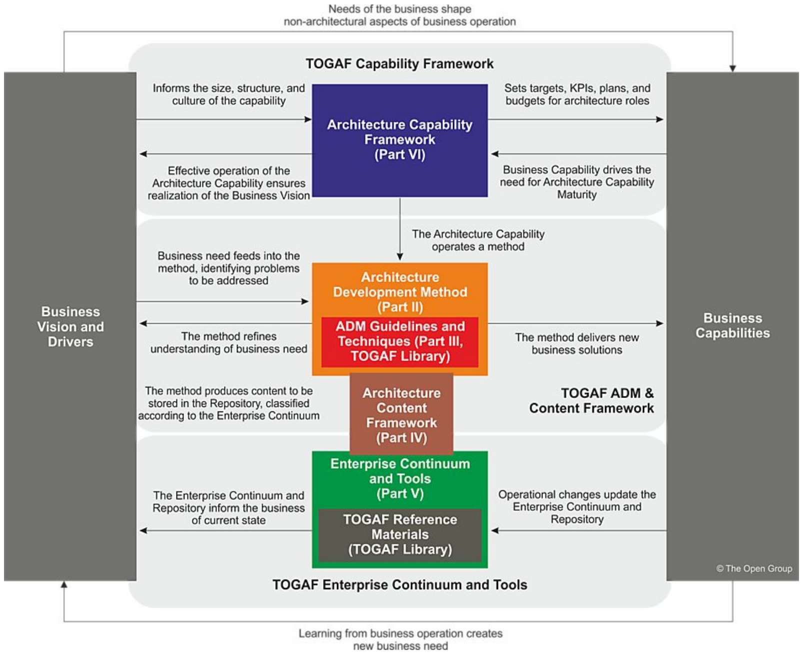
Operating a mature Architecture Capability generates substantial architectural output. The repository organises these through a formal taxonomy, categorising assets like models, diagrams, and documentation.

Management and Utilisation

Effective management requires dedicated processes and tools for content storage, retrieval, and governance, enabling stakeholders to access information supporting decision-making.

Framework Overview

TOGAF Capability Framework – Bridging Business Vision and Capabilities



Aligning Vision with Capabilities

TOGAF helps organisations align their strategic vision with necessary capabilities, ensuring architecture supports business objectives.

Defining the 'Why' and 'How'

Strategic drivers define 'why' an organisation aims for certain goals, while capabilities outline 'how' to achieve them.

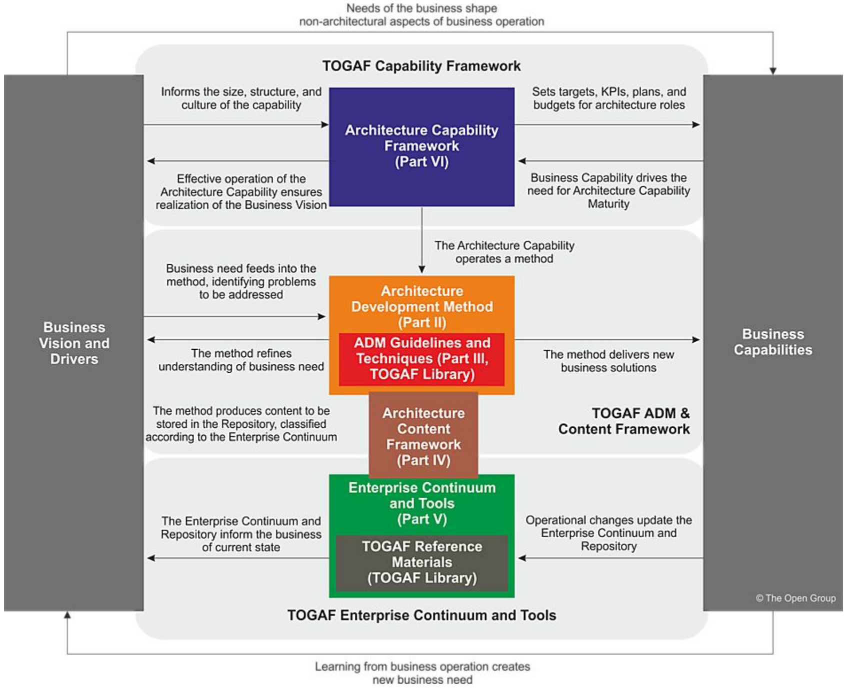
Fueling Organisational Transformation

The framework identifies capability gaps, aligns strategies, and supports change management for achieving business transformation.

Continuous Capability Development

Ensures ongoing assessment, planning, and governance to optimise and sustain capabilities over time.

TOGAF



Needs of the business shape
non-architectural aspects of business operation
TOGAF Capability Framework
Informs the size, structure, and culture of the capability
Architecture Capability Framework (Part VI)
Sets targets, KPIs, plans, and budgets for architecture roles
Effective operation of the Architecture Capability ensures realization of the Business Vision
Business Capability drives the need for Architecture Capability Maturity
The Architecture Capability operates a method
Business need feeds into the method, identifying problems to be addressed
Architecture Development Method (Part II)
ADM Guidelines and Techniques (Part III, TOGAF Library)
The method refines understanding of business need
The method delivers new business solutions
Business Vision and Drivers
Business Capabilities
The method produces content to be stored in the Repository, classified according to the Enterprise Continuum
Architecture Content Framework (Part IV)
TOGAF ADM & Content Framework
Enterprise Continuum and Tools (Part V)
TOGAF Reference Materials (TOGAF Library)
The Enterprise Continuum and Repository inform the business of current state
Operational changes update the Enterprise Continuum and Repository
© The Open Group
TOGAF Enterprise Continuum and Tools
Learning from business operation creates
new business need

TOGAF Capability Mapping

20 minutes
5 min class discussion

https://docs.google.com/document/d/1f5uQrjNHnyaNPiXI3G_AQMt-1zLIT29pmxTrbax_g50/edit?usp=sharing

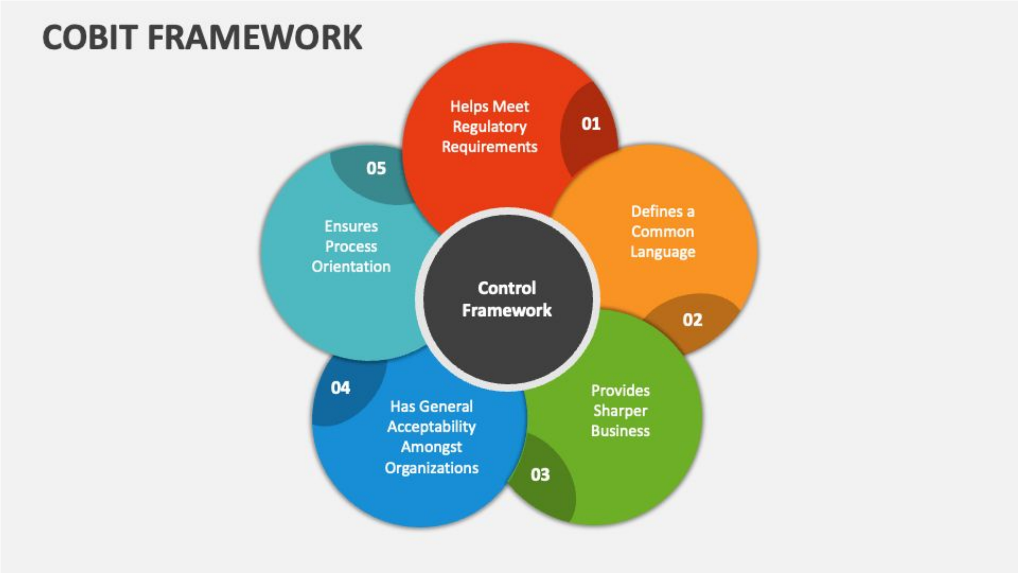
COBIT

COBIT (Control Objectives for Information and Related Technologies) is a globally recognized framework for IT governance and management. Developed by ISACA, COBIT ensures information and technology (I&T) supports enterprise goals, optimizes risks, and provides value.

- Framework that provides a set of best practices for IT governance and risk management
- Supports organizations to ensure their business goals are met through effective IT governance
- It encompasses:
  - Control Objectives:
    - Control objectives structure.
    - Relationship with business processes.
  - Alignment with Business Objectives:
    - How COBIT contributes to achieving organizational objectives.
    - Approach to risk management.

COBIT

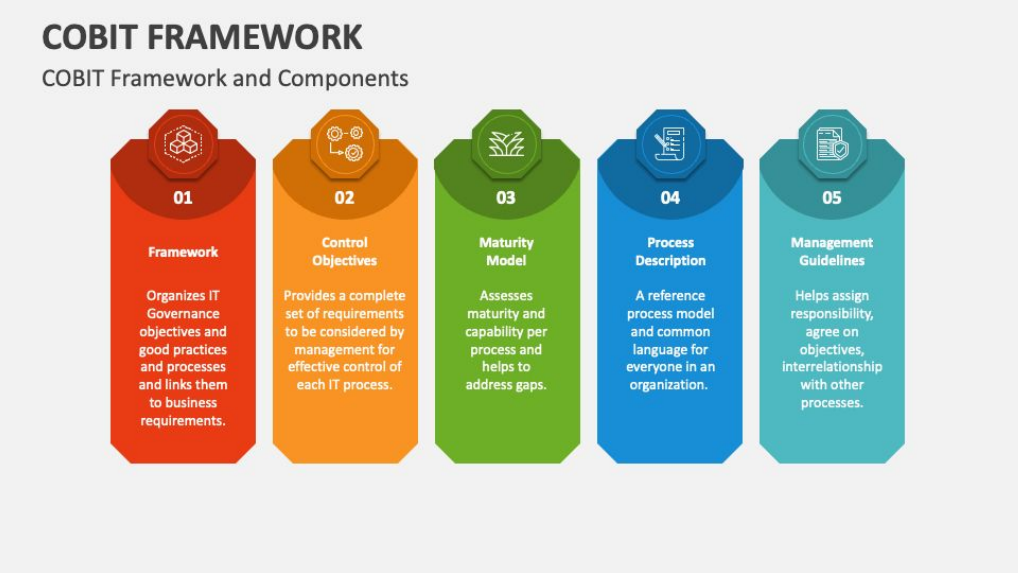
COBIT provides the principles, practices, analytical tools, and models to achieve this alignment and governance.



COBIT

COBIT's key components include Control Objectives, Maturity Models, Processes and Management Guidelines.

Main focus: balancing benefits realization, risk optimization, and resource optimization.

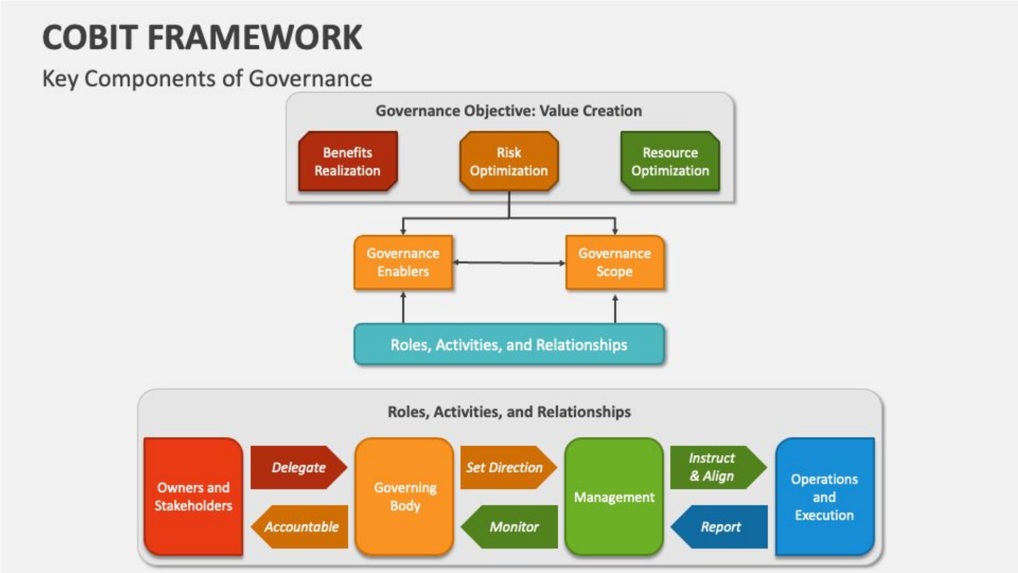


COBIT

COBIT facilitates the alignment of IT with business strategies by providing a framework that translates business objectives into IT goals and metrics.

It helps stakeholders understand their responsibilities toward governance and ensures IT-related decisions are made in alignment with the organization's strategic objectives.

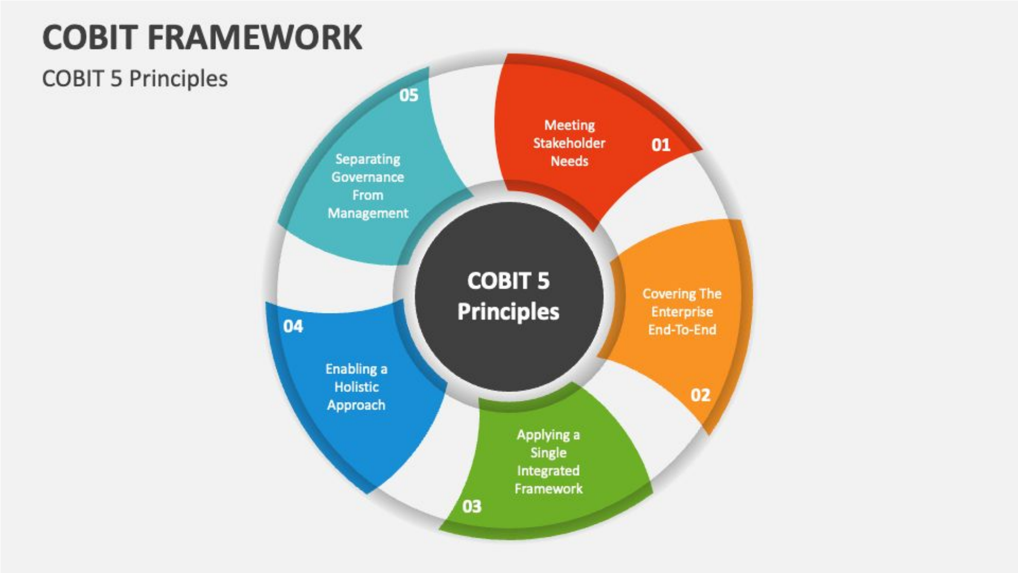
It keeps IT steered by the Owners and Stakeholders.



COBIT

COBIT principles enhance an organization's ability to govern and manage IT resources effectively.

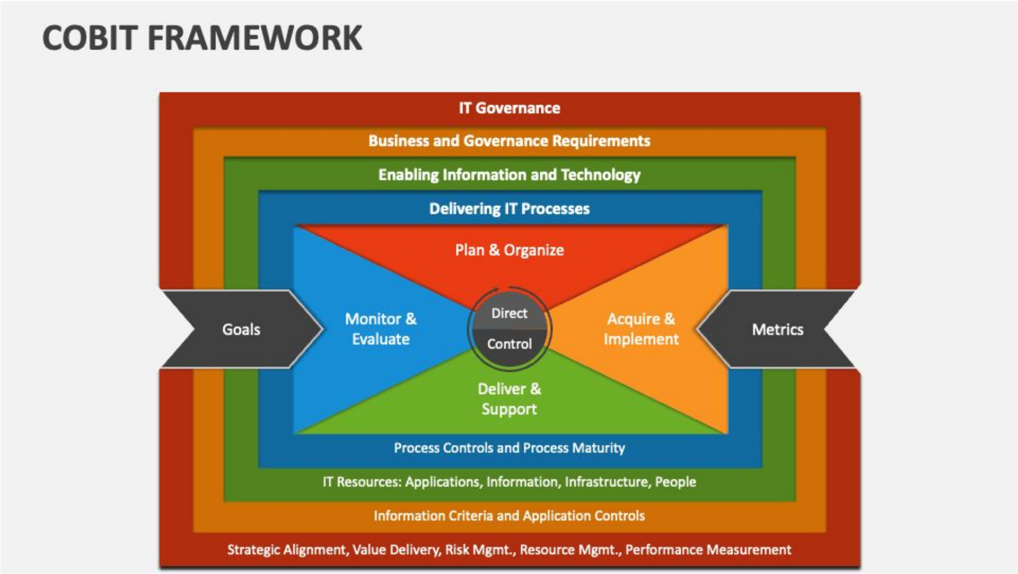
By aligning IT strategies with business objectives, ensuring comprehensive coverage across the enterprise, integrating with other frameworks, adopting a holistic approach, and clearly distinguishing between governance and management, organizations can achieve strategic goals, manage risks effectively, and optimize resource utilization.



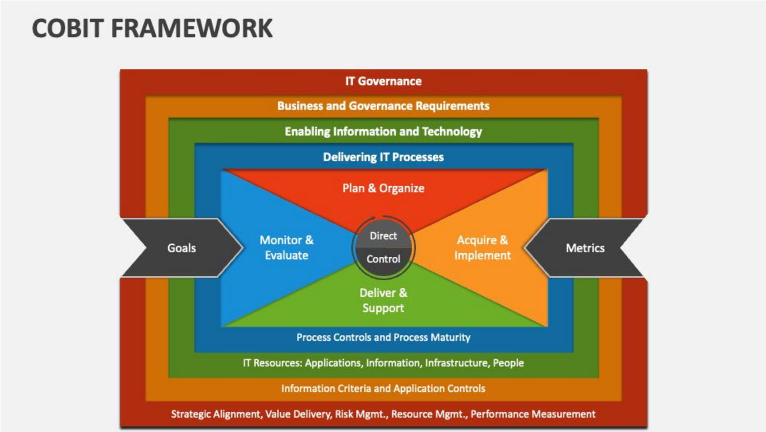
COBIT

Implementing COBIT involves assessing the current state of IT governance, identifying gaps, and applying COBIT principles to develop a tailored governance framework that meets the organization's unique needs.

This includes establishing clear policies, roles, and responsibilities, and setting measurable objectives.



COBIT
Domains



1. **Governance and Framework:**

Focuses on the overall governance structure, ensuring that IT governance is aligned with business goals and delivers value while balancing risk and resource use. It sets the stage for the entire governance framework, defining and establishing the components necessary for effective governance, including policies, processes, organizational structures, and ensuring compliance with relevant laws and regulations.

2. **Strategy and Planning (plan & organize):**

Strategy Management: Involves defining the direction, planning, and tactics necessary to achieve business objectives through IT. It covers the development of IT strategy in alignment with business strategy, ensuring that IT delivers the intended benefits.

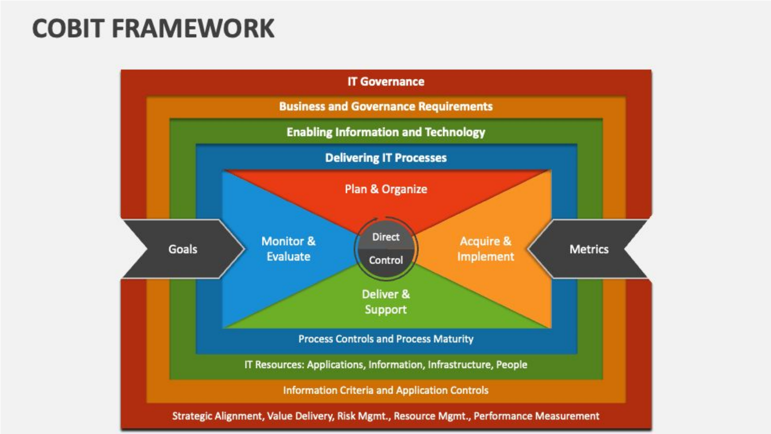
Enterprise Architecture Management: Deals with the structuring of IT infrastructure and operations in a way that supports the organization's strategy. This includes the definition of IT architecture and ensuring it aligns with business processes.

3. **Build and Acquire (Acquire & implement)**

Program and Project Management: Focuses on the management of IT projects and programs, ensuring they are delivered on time, within budget, and according to specifications while delivering the expected benefits to the business.

IT Solution Delivery and Support: Covers the development, procurement, and implementation of IT solutions that meet business requirements. It ensures that IT services are delivered effectively and support business operations.

COBIT
Domains



4. **Deliver and Support**

Service Delivery Management: Ensures that IT services are delivered in accordance with agreed-upon levels of service. It covers the management of IT service delivery processes to meet business needs reliably and efficiently.
Risk Management: Involves identifying, assessing, and managing IT-related risks to ensure they are within acceptable levels. It ensures that the organization's risk exposure is managed and that business continuity is maintained.
Resource Optimization: Focuses on optimizing the use of IT resources, including infrastructure, applications, information, and people. It ensures that IT resources are used efficiently and effectively to support business objectives.

5. **Monitor and Evaluate**

Performance and Conformance Monitoring and Reporting: Deals with the monitoring and reporting of IT performance against agreed-upon metrics and compliance with policies, standards, laws, and regulations. It ensures that IT is delivering value to the business and meeting its obligations.
Internal Control and Compliance: Focuses on the establishment and maintenance of internal controls to ensure reliability and integrity of information, compliance with policies and regulations, and effective and efficient operations.
Governance of IT Investments: Ensures that IT investments are aligned with business goals and deliver the expected return on investment. It involves the governance of IT investment decisions and the monitoring of IT investment performance.

COBIT

COBIT stands as a critical framework for aligning IT operations with business objectives, ensuring effective governance, and fostering a culture of continuous improvement and compliance.

Risk analysis and management

Risk Analysis and Management watches for issues concerning data integrity and maintaining operational security and stability. This process involves:

- **Identifying potential threats** that could impact IT systems, from cyber-attacks and data breaches to system failures and natural disasters.
- **Assessing the risks** by evaluating the likelihood of these threats occurring and their potential impact on the organization. This assessment helps in understanding the severity and extent of each risk.
- **Prioritizing risks** based on their severity and impact, ensuring that resources are allocated effectively to address the most critical vulnerabilities first.

Risk Management in COBIT and TOGAF

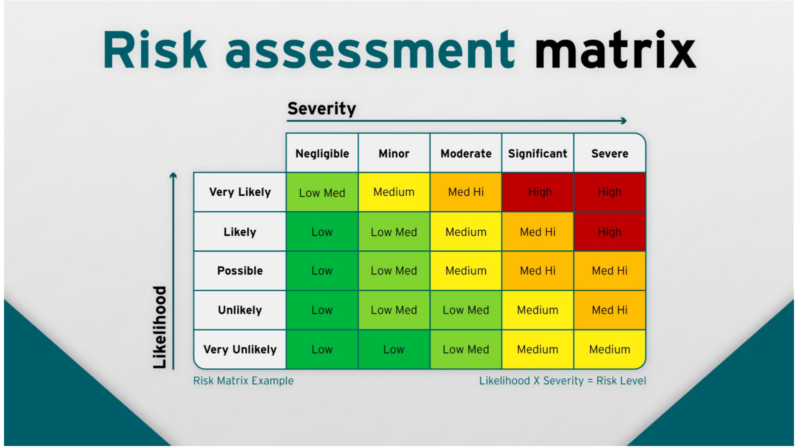
COBIT's Risk Management: Emphasizes aligning IT risk management practices with **business objectives**, ensuring an integrated approach to managing IT-related risks. COBIT provides a framework for establishing **controls and measures to mitigate identified risks**, monitoring and continually improving risk management practices.

TOGAF's Risk Management: Focuses on identifying and managing risks throughout the enterprise architecture development process, using the ADM. **TOGAF encourages proactive risk identification and mitigation strategies** to be embedded within the architecture planning and implementation phases.

Risk appetite

Risk appetite refers to the amount and type of risk an organization is willing to accept in pursuit of its objectives. Establishing clear risk appetite strategies is crucial for guiding decision-making and risk management practices. Key components of developing these strategies include:

- **Defining Risk Appetite**: the organization's willingness to take on risk, considering both quantitative and qualitative factors. It involves setting clear thresholds for different types of risks, such as operational, financial, and reputational risks.
- **Aligning with Business Objectives**: Risk appetite aligns with the organization's strategic goals and objectives. This alignment helps in prioritizing resources and efforts towards managing risks that could impede achieving these goals.
- **Developing Risk Tolerance Levels**: While risk appetite defines the overall willingness to accept risk, risk tolerance specifies acceptable variations in outcomes. Establishing tolerance levels for key performance indicators helps in monitoring and controlling risks.



Risk management strategies

- **Avoid:** Eliminating the risk by discontinuing the activities that generate it. This approach is suitable for high-level risks that surpass the organization's risk appetite and could significantly impact business objectives. *(not that much related to software)*
- **Mitigate:** Implementing measures to reduce the likelihood or impact of the risk. Mitigation strategies include enhancing security protocols, improving operational procedures, or adopting new technologies. This approach is commonly used for risks that are within the organization's risk appetite but still require management to ensure minimal impact.
- **Transfer:** Shifting the risk to a third party, such as through insurance or outsourcing. This strategy is particularly relevant for risks where the organization might not have direct control or where transferring the risk is more cost-effective than in-house mitigation.
- **Accept:** Recognizing that the risk is within the organization's risk appetite and deciding not to take specific actions to alter its likelihood or impact. This decision is often reserved for low-level risks or when the cost of other strategies outweighs the benefit.

Risk Matrix Analysis

20 minutes
5 min class discussion

https://docs.google.com/document/d/1b6QWdQbBqg0w9FjBuSexSOoAT1NzJf-J-3ApBrzdOEI/edit?usp=sharing

Integration

Connecting IT Governance and Software Architecture

Strategic Alignment in Software Architecture

- Software architecture bridges strategic objectives with technology solutions.
- Adhering to frameworks like TOGAF supports business goals.
- Ensures design aligns with organisational standards.

Risk Management Through Architectural Frameworks

- Facilitates risk management with secure and resilient design guidelines.
- Incorporates security, scalability, and compliance at design level.
- Addresses risks in foundational IT structure.

The Role of Software Architecture in Organisational Success

Alignment with Business Strategy

- Software architecture ensures systems align with business strategy and adapt to changes.
- Defines interactions, data flow, and integration of services for competitive advantage.
- Facilitates innovation by allowing quick adaptation to new demands and technologies.

Benefits of Proper Alignment

- Enhances innovation and agility, enabling quick market adaptation and maintaining competitiveness.
- Improves operational efficiency by reducing redundancies and enhancing interoperability.
- Focus on architecture ensures systems meet performance expectations and scale with growth.

Integration

Integrating Software Architecture with IT Governance and Architectural Frameworks

01 **Ensures Alignment**

Aligns software architecture practices with frameworks like TOGAF and governance models like COBIT to support strategic vision and comply with governance policies.

02 **Facilitates Communication**

Architectural frameworks provide a common language and standards for communication among stakeholders, ensuring clear understanding of architecture and business implications.

03 **Promotes Best Practices**

Encourages use of industry best practices in software design and development, leading to higher quality systems aligned with business needs.

Skills

Soft Skills for Software Architects

Beyond technical expertise, software architects must possess a robust set of soft skills to effectively lead projects and drive organisational success. These skills include communication, leadership, negotiation, problem-solving, and adaptability. Mastering these soft skills enables architects to bridge the gap between technical teams and business stakeholders, ensuring architectural visions are aligned with business goals and effectively implemented.

Skills Enhancement

Mastering Communication and Leadership

Effective Communication is critical for articulating technical concepts to non-technical stakeholders and ensuring clear understanding within development teams.

Leadership involves inspiring and guiding teams towards achieving architectural goals, managing conflicts, and fostering a collaborative environment.

Strategies for Improvement: Engage in cross-departmental projects, practice public speaking, and participate in leadership workshops or mentorship programmes.

Skills Development

Enhancing Negotiation and Problem-Solving Skills

Importance of Negotiation Skills

- Negotiation is key in achieving consensus among diverse stakeholder interests.
- Balances technical constraints with business needs effectively.
- Aids in fostering collaboration and mutual understanding amongst teams.

Strategies for Skill Improvement

- Attend negotiation training to improve consensus-building skills.
- Participate in design thinking workshops for innovative problem-solving.
- Engage in community hackathons to tackle real-world problems.

Professional Development

The Architect's CI/CD: Continuous Improvement, Continuous Deployment

Continuous Learning

Continuous learning and improvement are vital for staying ahead in the rapidly evolving field of software architecture.

Practising Regularly

Stay hands-on with coding and architectural design to maintain technical proficiency.

Participating in Open Source

Contribute to open source projects to gain exposure to diverse architectural patterns and practices.

Writing Blogs and Articles

Share knowledge and insights on architectural best practices, emerging trends, and personal experiences.

Teaching Others

Teach courses, lead workshops, or mentor juniors to refine your understanding and communication skills.

Networking

Join professional networks, attend conferences, and participate in forums to exchange ideas with peers.

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